Computer Networks

CSA-0710

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① a) Data link layer

The Data link layer is responsible for a transferring data between two directly connected devices over a physical network

Functions:

1.) Framing: Divides data into frames for transmission

2) Physical addressing: Use Mac addresses to identify devices on a local network

3) Error Detection: Detects transmission errors using methods such a CRL

4) Flow Control: Helps prevent a sender from overwhelming a receiver when supported by the link protocol

② 2. Transport Layer:

The transport layer provides communication between application running on different device

* Segmentation: Divides application data into smaller segments

* Error relovery: TCP detects missing data

(b)

Feature	Connection - oriented Service	Connection - Service
Protocol	TCP	UDP
Connected setup	Required	Not required
Relaiability	Supports relaible delivery	No delivery gaurantee
Acknowledgement	uses Acknowledgement	No built in acknowledgements
Data ordering	Maintains ordered delivery	No ordering gaurantee
Speed and overhead	Generally more overhead	Generally low overhead
Example	File transfer, web browsing using HTTS over TCP	online gaming, ons queries live streaming Using UDP

TCP is prefered when relaible, ordered delivery is required, whereas UDP is usefull when low latency and low overhead are important

(2)

a) Encapsulation:

It is a process of adding protocol information to data as it moves down the OSI layer of the source device

OSI layer	Process	Data unit
Application	Creates application data	Data
Presentation	Handles translation, Encryption or the Compression as needed	Data
Session	Manages Communication session	Data
Transport	Adds transport header such as TCP and UDP	Segment
Network	Adds MAC addressing and frame checking information	Packet
Data link	Adress source	frame
Physical	Transmits bits over the medium	Bits

⑤ Classfull IP adressing:

In classfull adressing, IPV4 adresses where divided into fixed classes A, B, C, D and E

class	First octet-range	Default Mask	Purpose
A	1-126	255-0.0.0(/8)	Large Network
B	128-191	255-255-0.0(/16)	Mediumsize networks
C	192-223	255-255-0(/24)	Small networks
D	224-239	Not applicable	multicast
E	240-255	Not applicable	Reverse

CIDR improves IPV4 adress utilization and routing efficiency by using flexible network prefix lengths, smaller organization can receive smaller adress blocks

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Given

* Network address = 172.16.0.0 /16
* Required subnets = At least 6

Step1:

$2^n \geq$ Required subnets

For 6 Subnets

$2^2 = 4$

$2^3 = 8$

∴ 3 bits must be borrowed

Step2:

Prefix = /16

Borrowed = 3

New prefix = 16 + 3 = 19

∴ New CIDR prefix is /19 255.255.224.0

∴ CIDR notation : /19

Dotted decimal subnet mask : 255 : 255.224.0